

Eshat Jahan Arifa

B.Sc. in Biomedical Engineering

01960298116 | eshatarifa@gmail.com | [LinkedIn](#) | [Website](#) | [Google Scholar](#) | [ResearchGate](#)

Education

Year	Degree & Institute	CGPA
2026	Bachelor of Science in Biomedical Engineering, Chittagong University of Engineering & Technology (CUET), Bangladesh	3.62/4.00

Research Interests

Biomedical Signal Processing, Biomaterials and Drug Delivery Systems, Electrochemical Sensing, Medical Image Analysis, Neural Engineering and Brain-Computer Interfaces

Undergraduate Thesis

Development and Characterization of Natural Polymer-Based *Calendula officinalis* Loaded Hydrogel for Wound Dressing Application (*manuscript in preparation*) Nov 2025 – Jun 2026
Supervisor: Ayesha Binth Humayun

- Developed and characterized gelatin–sodium alginate hydrogels incorporated with *Calendula officinalis* extract for potential wound-dressing application.
- The composition shows better results than the existing samples.

Research Publications

- Eshat Jahan Arifa** and Ayesha Binth Humayun, “Development and Characterization of Natural Polymer-Based *Calendula officinalis* Loaded Hydrogel for Wound Dressing Application.” (*Manuscript in preparation*).
- Md. Taisirul Muktdi, Tanzeem Tahmeed Reza, Mohammed Tawshif Hossain, **Eshat Jahan Arifa**, and Md. Fazlul Karim Khondakar, “Spectrogram-driven Emotion Detection from Electroencephalogram,” in *2025 4th International Conference on Electrical, Computer and Communication Engineering (ECCE)*, CUET, Chattogram, Bangladesh, IEEE, 2025. DOI: [10.1109/ECCE64574.2025.11013815](https://doi.org/10.1109/ECCE64574.2025.11013815). (Published)
- Tanzeem Tahmeed Reza, Sharafat Hossain, and **Eshat Jahan Arifa**, “Novel Task-Specific Features Extraction and Comparative Evaluation for Parkinson’s Detection from Digital Pen Signal Data,” in *2025 IEEE International Conference on Biomedical Engineering, Computer and Information Technology for Health (BECITHCON)*, IEEE, 2025. DOI: [10.1109/BECITHCON69222.2025.11504012](https://doi.org/10.1109/BECITHCON69222.2025.11504012). (Published)

Work Experience

Research Assistant, Electrochemical Lab, CUET Jul 2026 – Present
Supervisors: Dr. Sumit Majumder and Dr. Arup Kumer Roy

- Conducting electrochemical measurements and analyzing responses for hydrocortisone detection (work in progress).

Internship Experience

Intern, Tradevision LTD Apr 2025 – May 2025

- Gained practical exposure to the operation and working principles of CT scanners, CT simulators, ultrasonography systems, dialysis machines, and ophthalmology equipment.
- Observed the operation of ICU facilities, cyclotron systems, medical gas pipeline systems, and hospital waste management practices.
- Developed practical understanding of biomedical equipment, hospital infrastructure, and healthcare technology management.

Projects

Consumer Choice Prediction Using EEG Signals 2025

- Developed an EEG-based machine learning framework to classify consumer decisions (Buy vs. No-Buy) using neural activity patterns from the NeuMa dataset.

Skin Lesion Boundary Detection Using Classical Image Processing 2024

- Developed a medical image segmentation pipeline to detect skin lesion boundaries from dermoscopic images using the PH2 dataset.

Movement Pattern Recognition and Phase Detection Using Wearable Sensors 2024

- Built a wearable motion classification system to identify standing, walking, and falling phases.

- Designed a real-time biomedical monitoring architecture for capturing and wirelessly transmitting heart and lung sound data.

- Developed a low-cost embedded healthcare solution for detecting moisture conditions in bedridden patients.

Relevant Coursework

Biomaterials, Prosthetics and Orthotics	Biomedical Signals and Systems
Biomedical Signal Processing	Bioelectricity
Biomedical Instrumentation	Brain and Neuroengineering
Biosensors and Biochips (Elective I)	Bioinformatics (Elective III)

Technical Skills

Programming & Libraries:	Python, C/C++, NumPy, Pandas, SciPy, Scikit-learn, Matplotlib
Biomedical Computing & Laboratory:	MATLAB, Simulink, EEGLAB, AutoDock Vina, PyMOL, Signal Processing, Medical Image Analysis, Machine Learning, Cyclic Voltammetry, Potentiostat Operation, Hydrogel Fabrication, In-vitro Testing
Documentation:	LaTeX / Overleaf, Microsoft Word, Excel, PowerPoint

Achievements & Certifications

- Galactic Problem Solver, NASA International Space Apps Challenge (2025, 2024)
- Champion, BioOlympiad – Department of Biomedical Engineering, CUET (2025)
- Champion, Idea Contest – Department of Biomedical Engineering, CUET (2025, 2022)
- Finalist, Face the Case Competition – IEEE CUET Student Branch (2025, 2023)
- Participant, 3-Minute Thesis (3MT) Competition, SciBlitz 2.0 – IEEE CUET Student Branch (2026)

Extracurricular & Volunteer Activities

- Joint General Secretary, IEEE EMBS CUET Student Chapter (August 2025 – August 2026)
- Assistant General Secretary, BMES CUET (August 2025 – August 2026)

Language Proficiency

Medium of Instruction:	English
IELTS:	Scheduled to appear in October 2026

References

Dr. Sumit Majumder Assistant Professor, Department of Biomedical Engineering, CUET Chittagong University of Engineering & Technology Email: s.majumder@cuet.ac.bd	Dr. Md. Fazlul Karim Khondakar Assistant Professor, Department of Biomedical Engineering, CUET Chittagong University of Engineering & Technology Email: fazlul@cuet.ac.bd
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